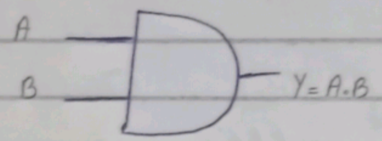
Lab Task # 11

Question:

Draw Logic circuit and truth table.

AND Gate:

Output is high (1) only if all inputs are high

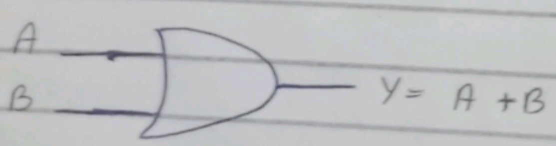


AND

A	B	Y = A.B
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate:

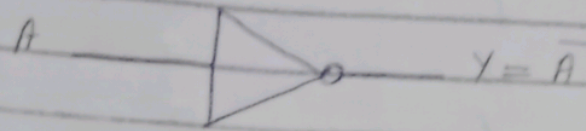
Output is high (1) if at least one input is high.



A	B	Y = A+B
0	0	0
0	1	1
1	0	1
1	1	1

NOT Gate (Inverter) :

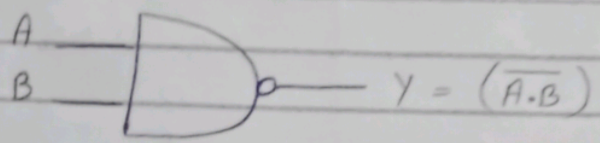
Output is inverse of input.



A	$Y = \overline{A}$
0	1
1	0

NAND Gate (NOT-AND) :

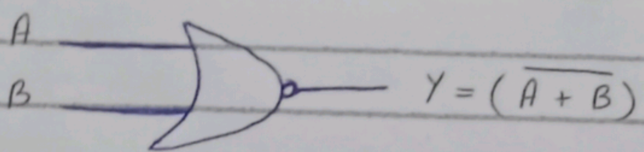
Output is low (0) only if all inputs are high (opposite of AND).



A	B	$Y = \overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate (NOT-OR) :

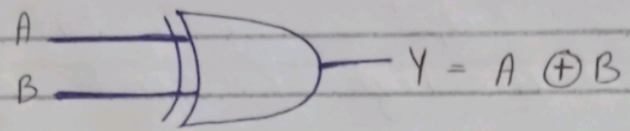
Output is high (1) only if all inputs are low (opposite of OR)



A	B	$Y = (\overline{A+B})$
0	0	1
0	1	0
1	0	0
1	1	0

XOR Gate (Exclusive OR) :

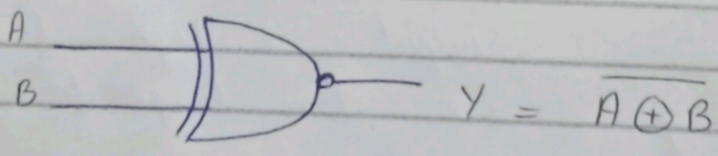
Output is high 1 only if inputs are different.



A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

XNOR Gate

Output is high (1) only if all inputs must be same.



A	B	Y = $\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1